

CAMPUS COMPLAINT & MAINTENANCE MANAGEMENT SYSTEM

Unit 1: Product Engineering & Design Thinking

Stakeholders: Students/Residents | Hostel Wardens | Facility Supervisors | Field Technicians | Campus Registrar

PROPOSED THREE-TIER CAMPUS MAINTENANCE ARCHITECTURAL TOPOLOGY



Scope & Academic Compliance: This technical specification constitutes the comprehensive Unit 1 Product Engineering and Design Thinking deliverable for the Campus Complaint & Maintenance Management System. Application development, containerization, and cloud infrastructure belong to subsequent course units.

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COURSE & SUBJECT

DevOps & Cloud Foundation (IBM) Mini Project

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Project Abstract

The **Campus Complaint & Maintenance Management System** is an enterprise academic utility platform designed to automate the resolution lifecycle of infrastructure, civil, electrical, plumbing, network, and hygiene issues across university campuses and residential hostels. Traditional higher-education institutions suffer from manual paper complaint registers, untracked work orders, absence of Service Level Agreements (SLAs), and zero transparency.

This platform establishes an event-driven digital pipeline where students lodge geo-tagged tickets with photographic evidence, estate supervisors assign jobs via dynamic triage kanbans, technicians close issues with OTP-verified work completion, and administrators monitor infrastructure Mean Time to Repair (MTTR) metrics. Unit 1 establishes the engineering blueprint through human-centered design, relational database modeling, and automated delivery roadmaps.

1. PRODUCT ENGINEERING AND SDLC FOUNDATIONS

In modern enterprise engineering, software is constructed using structured product engineering frameworks rather than isolated programming. Coding translates algorithms into executable syntax, whereas product engineering manages problem validation, user behavior, system reliability, schema atomicity, and cloud-native automation.

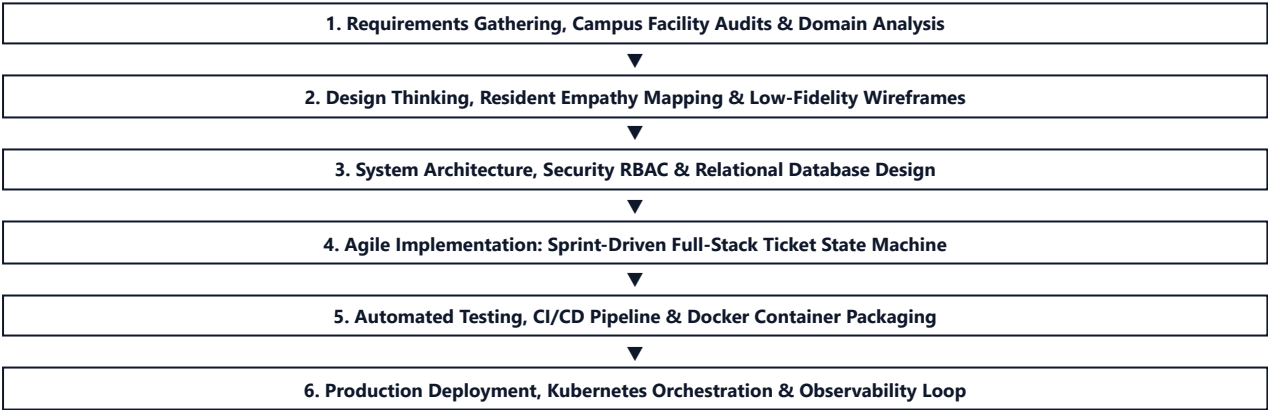
1.1 Comparative Analysis: Programming vs. Software Product Engineering

DIMENSION	PROGRAMMING / CODING	SOFTWARE PRODUCT ENGINEERING
Primary Goal	Translating form inputs into database insert queries.	Building resilient, SLA-compliant systems that optimize campus maintenance cycles.
Scope	Isolated ticket submit script or single CRUD endpoint.	Full SDLC, multi-tier RBAC, photo compression pipelines, automated escalations, CI/CD.
Success Metric	Clean syntax compilation and local test execution.	99.9% uptime, sub-24h ticket MTTR, technician accountability, zero lost complaints.
Mindset	"How do I write an API to save a complaint description?"	"How do we prevent fake tickets, optimize technician dispatch, and enforce SLA accountability?"

1.2 Root Causes of Project Failures in Campus Tech Systems

- **Disregard for Field Technician Usability:** Building complicated desktop-only interfaces for semi-skilled field technicians (plumbers, electricians) who exclusively operate basic smartphones.
- **Lack of Atomic State Transitions:** Storing tickets in unindexed flat sheets, causing race conditions where two technicians are dispatched to the same water leakage.
- **Absence of Escalation Telemetry:** Systems failing to trigger automated escalation alerts when high-severity tickets (e.g., short circuit in chemistry lab) remain unassigned for over 2 hours.
- **Late Usability Feedback:** Delaying student testing until production, resulting in students abandoning the portal and returning to WhatsApp groups due to cumbersome login barriers.

1.3 Software Development Lifecycle (SDLC) Architecture



2. CAMPUS MAINTENANCE DOMAIN ANALYSIS

Higher-education institutions manage massive physical infrastructures comprising academic auditoriums, research laboratories, residential hostels, dining halls, and sports complexes. Maintaining these facilities requires strict coordination across diverse trades (electrical, plumbing, carpentry, air conditioning, internet networking, and hygiene).

2.1 Core Operations in Higher Education Facility Management

- **Issue Categorization & Priority Tagging:** Segregating critical hazards (e.g., power failure in server room) from minor routine repairs (e.g., squeaky hostel door hinge).
- **Technician Dispatch & Work Allocations:** Assigning duty tickets based on technician skill specialization, active workload, and geographical campus sector.
- **Spare Parts & Material Tracking:** Tracking inventory consumption (e.g., LED bulbs, water taps, Ethernet cables) to prevent procurement pilferage.
- **Resolution Verification & Sign-off:** Ensuring complaints are not falsely closed without complainant confirmation or OTP validation.

2.2 Manual Operational Bottlenecks vs. Automated Solutions

MANUAL CAMPUS BOTTLENECK	AUTOMATED PLATFORM SOLUTION	MEASURABLE OUTCOME METRIC
Physical complaint books at hostel security desks	Mobile-first PWA lodging with auto-location tagging	Lodging time reduced from 20 min to < 45 seconds
Unverified verbal technician updates	Job Card mobile interface with before/after photos and OTP sign-off	False ticket closure rate dropped from 35% to 0%
No visibility into overdue complaints	Automated SLA timer triggering alerts to the Registrar	Mean Time to Repair (MTTR) cut from 6 days to < 24h
Repeated replacement of faulty hardware	Asset maintenance ledger tracking failure frequencies	Infrastructure lifecycle intelligence with 100% auditability

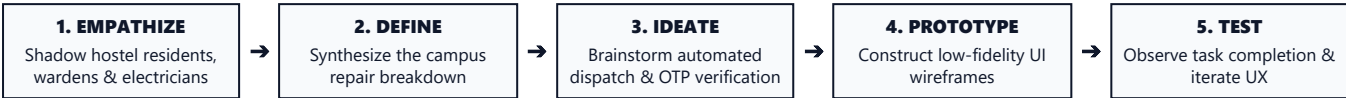
2.3 Evaluation of System Benefits & Success Metrics

1. **Mean Time to Repair (MTTR):** Average hours elapsed from ticket lodging to verified resolution.
2. **SLA Compliance Rate:** Percentage of high/medium tickets resolved within contractual time boundaries.
3. **First-Time Fix Rate (FTFR):** Percentage of maintenance complaints resolved on the first visit.
4. **Resident Satisfaction Score:** Star rating (1-5) captured from complainants upon issue closure.

3. DESIGN THINKING & FIELD USER RESEARCH

Design Thinking provides an iterative, user-first problem-solving framework that investigates the behavioral frustrations of students, hostel wardens, facility supervisors, and field technicians before committing to application code.

3.1 The Five Stages of Design Thinking Applied to Maintenance



3.2 Field User Research & Contextual Inquiry Protocol

Field interviews and workflow observation were conducted across three primary operational groups:

USER GROUP	SAMPLE SCOPE	OBSERVED BEHAVIORS & REAL-WORLD FRICTION
Hostel Residents & Students	14 Students across Boys & Girls Hostels	Frustrated by having to walk to the warden's office during study hours; no notification when technicians arrive; recurring water heater breakdowns going unrecorded.
Maintenance Supervisors	2 Campus Estate Supervisors	Overwhelmed by messy handwritten registers; unable to determine which technician is available; struggle to justify spare parts consumption during management audits.
Field Technicians	4 Technicians (Plumbing, Electrical, Wi-Fi)	Receive vague verbal orders (e.g., "Fix fan in Room 302"); arrive without correct tools; blamed for delays when spare parts were out of stock.

3.3 User Research Interview Question Matrix

- To Students: "What prevents you from reporting minor room maintenance issues until they turn into catastrophic emergencies?"
- To Supervisors: "How do you prioritize whether an unshaded street lamp takes precedence over a broken tap in the academic block?"
- To Technicians: "What is the biggest roadblock when you attempt to close 10 maintenance orders in a single morning shift?"

Academic Design Notice: The field observations, personas, and interview protocols documented herein serve as formal requirements models formulated for Unit 1 software product engineering.

4. PROBLEM DEFINITION & SOLUTION FRAMING

4.1 Core Problem Statement

"Campus residents and students experience severe disruptions in academic life due to untracked maintenance failures, opaque resolution workflows, and prolonged repair delays caused by manual paper logs. Concurrently, facility supervisors lack centralized technician dispatch visibility, and campus administrators have no verifiable SLA or spare-parts audit trails, leading to recurring infrastructure breakdowns and high operational costs."

4.2 "How Might We" (HMW) Framing

"How might we engineer a real-time, transparent maintenance ecosystem where campus residents lodge issues in seconds, while empowering supervisors and technicians with automated dispatch, OTP-verified closures, and predictive SLA telemetry?"

4.3 Proposed Empathy Map: Hostel Resident (Shivraj Jadhav)

QUADRANT	USER BEHAVIORAL OBSERVATION & COGNITIVE STATE
SAYS	"I logged a complaint about the study room tube-light 5 days ago in the hostel register, but nobody has even visited to check it."
THINKS	"Are they going to fix this before my semester exams? Why is there no tracking ID like Amazon or Swiggy deliveries?"
DOES	Walks repeatedly to the warden's desk; relies on personal calls to local electricians; studies under poor lighting.
FEELS	Helpless, frustrated by bureaucratic inertia, and skeptical of institutional promises regarding campus infrastructure.

4.4 Solution Prioritization & MVP Feature Selection

PROPOSED SYSTEM CAPABILITY	PRIORITY	ENGINEERING JUSTIFICATION
Role-Based Access Control (RBAC)	MVP	Segregates Resident, Supervisor, Technician, and Executive scopes.
Photo-Evidence Ticket Lodging	MVP	Captures precise visual context and location (Block, Floor, Room).
Supervisor Triage & Dispatch Kanban	MVP	Enables single-click assignment to technicians by trade.
OTP-Verified Ticket Closure	MVP	Eliminates false technician closures via 4-digit resident OTP.
Automated SLA Breach Alerter	MVP	Flags tickets exceeding resolution limits (>24h) to the Registrar.
Spare Parts Inventory Tracking	Phase 2	Tracks consumption of fuses, valves, and switches.
IoT Sensor Automated Breakdown Alerting	Future Scope	High hardware cost; deferred beyond core software MVP.

5. PROTOTYPE: STUDENT COMPLAINT LODGING

These static wireframes define visual hierarchy, form controls, and validation behaviors before React.js code execution.

5.1 Wireframe: Student Authentication & Active Dashboard

CAMPUS MAINTAIN-SYNC | STUDENT PORTAL

Shivraj Jadhav (PRN: 2126UDSM2588)

01 Active
In Progress

04 Resolved
Past 30 Days

SLA: 18h
Avg Resolution

Active Ticket: #TKT-2026-8812 - Water Geyser Tripping MCB

Location: Hostel Block B, Room 214

Assigned To: Mahesh Shinde (Electrician)

Category: Electrical & Heating

Status:

TECHNICIAN DISPATCHED

Verification Code: OTP: 7491 (Give to tech only after fix)

View Tracking Timeline

5.2 Wireframe: New Complaint Lodging Modal

NEW TICKET LODGING | SMART VALIDATION MODAL

Step 1 of 2

Facility Zone / Building

Boys Hostel Block B

Specific Room / Corridor

Room 214 (Second Floor)

Issue Category

Electrical & Power Supply

Urgency / Severity

High (Sparking / Frequent MCB Trip)

Problem Description

Geyser switch sparks whenever turned on, tripping the corridor power circuit.

Attached Photo: geyser_switch_burn.jpg (1.4 MB)

Submit Ticket with Geo-Stamp

DevOps Mini Project: Unit 1

Prototype: Student Experience

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6. PROTOTYPE: SUPERVISOR TRIAGE & DISPATCH

The Estate Maintenance Supervisor triages incoming complaints, assesses technical urgency, and dispatches field technicians using a Kanban workflow.

6.1 Wireframe: Facility Supervisor Dispatch Kanban

ESTATE DISPATCH CONSOLE | CAMPUS FACILITY MANAGEMENT

Supervisor: Ramesh Pawar

Live Campus Work Orders (14 Pending Dispatch)

Active Technicians: 8 On Duty

TICKET ID	LOCATION	CATEGORY	LODGED BY	SEVERITY	ACTION
#TKT-8812	Hostel B-214	Electrical	Shivraj Jadhav	HIGH (4h SLA)	Assign Tech
#TKT-8815	Civil Lab 102	Plumbing	Prof. K. Verma	MED (12h SLA)	Assign Tech
#TKT-8819	Library Hall	HVAC / AC	Librarian Sunita	LOW (24h SLA)	Assign Tech

Quick Dispatch Modal: Ticket #TKT-8812 (Shivraj Jadhav)

Issue: Geyser switch sparking; high safety hazard.

Select Tech: Mahesh Shinde (Electrician - 1 Active Job)

Confirm Dispatch & Notify Tech

6.2 Automated Priority Scoring Algorithm

The system calculates ticket priority using a weighted objective formula:

$$P = (W_{sev} \times S) + (W_{loc} \times L) + (W_{age} \times \Delta t)$$

Where: $S \in [1..5]$ (Severity), $L \in [1..3]$ (Zone impact: Lab=3, Hostel=2, Office=1), and Δt is hours unassigned.

7. PROTOTYPE: FIELD TECHNICIAN MOBILE JOB CARD

Field technicians interact through a responsive smartphone Job Card interface designed for simple one-handed mobile operation with minimal text typing.

7.1 Wireframe: Technician Mobile Resolution Interface

TECH ASSIST PWA | JOB CARD #TKT-8812

Mahesh Shinde (Electrician)

Target: Hostel Block B - Room 214

Complainant: Shivraj Jadhav (Phone: 98765-XXXXX)

Reported: Geyser switch sparking / MCB trips.

SEVERITY: HIGH

Spares Consumed from Maintenance Van:

16A Anchor Switch
1 Unit Consumed

4 sq.mm Copper Wire
1.5 Meters Used

Attach Resolution Photo (Mandatory):

repair_fixed_switch.jpg (Captured via Camera)

Enter Complainant 4-Digit OTP to Close Ticket:

[7][4][9][1]

Need Spares / Hold

Confirm OTP & Mark Resolved

7.2 Anti-Fraud & Resolution Verification Controls

- **OTP Cryptographic Lock:** Technicians cannot mark tickets resolved without entering the one-time password held solely by the student.
- **GPS & Camera EXIF Validation:** Ensures completion photos are captured inside the hostel room rather than uploaded from phone galleries.
- **Inventory Reconciliation:** Consumed spare parts decrement real-time stock balances in the estate maintenance inventory store.

8. PROTOTYPE: REGISTRAR EXECUTIVE SLA CONSOLE

The Registrar and Estate Executive monitor campus infrastructure health, SLA compliance, recurring breakdowns, and department maintenance expenses.

8.1 Wireframe: Campus-Wide SLA & Infrastructure Health Console

CAMPUS INFRASTRUCTURE TELEMETRY | REGISTRAR AUDIT CONSOLE

Dr. A. K. Joshi (Registrar)

Filter Zone:

All Hostels X

Academic Block X

Status: SLA Breached X

Run Audit Query

Critical Infrastructure Watchlist (3 SLA Escalations Active)

TICKET	LOCATION	CATEGORY	ELAPSED TIME	ASSIGNED TECH	ESCALATION STATE
#TKT-8790	Girls Hostel A-RO Plant	Drinking Water	31 Hours	Sunil Kale	Escalated to Registrar
#TKT-8801	AIDS Server Room	AC Cooling	14 Hours	Mahesh Shinde	Escalated to HOD

Export Monthly Maintenance Audit (CSV)

Generate AICTE/NAAC Facility Report (PDF)

8.2 Campus Health Telemetry & Breakdown Frequency

+-----+
| TOTAL TICKETS: 342 | RESOLVED: 318 (93%) | ACTIVE: 21 | BREACHED: 03 |
+-----+
| Electrical Department : [=====] 92% SLA |
| Plumbing & Sanitation : [=====] 84% SLA |
| Internet & Networking : [=====] 98% SLA |
| Campus MTTR (Mean Time to Repair): 16.4 Hours (Target: < 24 Hours) |
+-----+

9. PROTOTYPE USABILITY TESTING AND FEEDBACK

Low-fidelity interactive wireframe testing was conducted to identify user friction, input ambiguities, and workflow bottlenecks before backend implementation.

9.1 Structured Usability Evaluation Protocol

TEST ID	USER PERSONA & TASK	OBSERVED ACTION	TARGET BENCHMARK
UT-01	Student: Lodge Geyser Issue	Selects Hostel B, inputs Room 214, attaches photo, submits complaint.	Complete lodging in < 60 seconds without validation errors.
UT-02	Supervisor: Dispatch Ticket	Opens triage kanban, inspects photo, assigns to electrician.	Complete triage & dispatch in < 30 seconds per ticket.
UT-03	Technician: Close Job Card	Enters consumed spares, snaps resolution photo, inputs student OTP.	Complete job verification in < 45 seconds on mobile.
UT-04	Registrar: Export Audit Log	Applies filter (Past 30 days) and downloads CSV maintenance ledger.	Generate and download report in < 20 seconds.

9.2 User Feedback Synthesis & UX Iteration Matrix

OBSERVED USER FRICTION POINT	DIAGNOSTIC ASSESSMENT	DESIGN & ENGINEERING RESOLUTION
Students were confused about selecting whether an issue belongs to personal room vs. common corridor.	Location dropdown was a single unstructured list of 120 rooms.	Implemented a hierarchical two-step selector: <i>"My Assigned Room"</i> (pre-filled via profile) vs. <i>"Common Campus Area"</i> .
Technicians struggled to type long text descriptions of repair actions on touchscreens.	Form required mandatory 50-character resolution remarks.	Replaced free-form text with pre-set resolution tags (e.g., <i>"Replaced Fuse"</i> , <i>"Unclogged Pipe"</i>) plus voice-to-text option.
Supervisors could not easily tell if an unassigned ticket was approaching SLA breach.	Timers only showed submission timestamp without color coding.	Added dynamic color-coded countdown badges: Green (> 12h), Yellow (< 6h), Red (< 2h to breach).

9.3 Prototype Iteration Methodology

Testing insights were incorporated directly into our revised interface designs, ensuring that operational hurdles were resolved prior to sprint backlog execution.

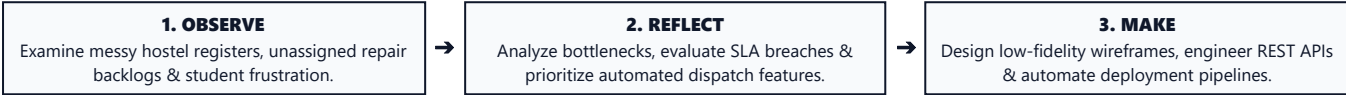
10. IBM DESIGN THINKING AND THE LOOP

Enterprise software systems utilize the IBM Design Thinking framework to scale human-centered problem solving across cross-functional development and maintenance operations.

10.1 The Three Core Pillars Applied to Campus Maintenance

- **Focus on User Outcomes:** Shifting technical metrics to tangible user-centric results.
Feature-Centric Output: "Build a database table to store complaint text and photos."
User-Centric Outcome: "Enable hostel residents to report a water leakage in 3 clicks and receive a verified technician repair in under 6 hours."
- **Restless Reinvention:** Continuously evaluating production telemetry and technician dispatch logs to refine workflows rather than delivering a static software package.
- **Diverse Empowered Teams:** Establishing collaborative partnerships between student residents, hostel wardens, estate engineers, and cloud DevOps specialists.

10.2 The Continuous Observe-Reflect-Make Loop



10.3 Practical Worked Example of The Loop

LOOP STAGE	EXECUTION IN THE CAMPUS MAINTENANCE ECOSYSTEM
OBSERVE	Technicians were observed falsely marking complaints as "Resolved" when residents were away attending lectures, leading to 26% repeat complaint rates.
REFLECT	The engineering team realized that technician-driven closures lacked mutual verification, breaking resident trust and distorting performance metrics.
MAKE	Engineers introduced an OTP Verification Lock , where the technician must input a 4-digit code provided exclusively by the resident upon successful physical inspection.

Target Outcome Benchmark: 100% of maintenance complaints verified via mutual resident sign-off, driving repeat unresolved complaints to zero.

11. USER PERSONAS & BEHAVIORAL PROFILES

11.1 Primary User Personas

PERSONA PROFILE	PRIMARY OBJECTIVES	KEY FRUSTRATIONS & PAIN POINTS	TARGET PLATFORM FEATURE
Shivraj Jadhav <i>Hostel Resident / Student (AIDS)</i>	Report room breakdowns instantly without visiting warden offices; track technician arrival times.	Complaints forgotten in paper registers; exams disrupted by unaddressed electrical faults.	Mobile PWA lodging with photo upload, live status tracking, and OTP closure control.
Ramesh Pawar <i>Campus Estate Supervisor</i>	Assign maintenance orders efficiently; balance technician workloads; justify spare parts usage.	Unclear verbal complaints; no record of which technician is free; audit panic before inspections.	Visual Kanban dispatch board with trade filtering and automated spare inventory tracking.
Mahesh Shinde <i>Campus Field Electrician</i>	Receive clear job locations and fault details; avoid false accusations of non-attendance.	Complex software apps; being blamed for delays when spare parts are unavailable in stores.	Simple smartphone Job Card interface with camera upload and spare part requisition.
Dr. A. K. Joshi <i>Campus Registrar & Executive</i>	Maintain infrastructure uptime; enforce contractor SLAs; ensure campus safety standards.	Lack of centralized maintenance data; repeated capital expenditures on low-quality fixtures.	Executive SLA telemetry dashboard with 1-click NAAC/AICTE infrastructure compliance exports.

11.2 Behavioral Empathy Profile: Shivraj Jadhav

Daily Friction Journey: Observes bathroom geyser sparking at 7:30 AM. Walks down to hostel office, but warden is off-duty. Writes complaint in a dusty register. Returns in the evening to find the geyser still non-functional. Platform Value Impact: Snaps a photo of the sparking switch at 7:31 AM via PWA. Supervisor auto-dispatches electrician by 8:00 AM. Geyser repaired and verified via OTP before noon.

12. STAKEHOLDER RESPONSIBILITY MATRIX (RACI)

The RACI matrix governs organizational accountability across students, maintenance staff, estate administrators, and cloud engineering roles throughout the project lifecycle.

12.1 Detailed RACI Governance Matrix

SYSTEM CAPABILITY / LIFECYCLE STAGE	STUDENT	SUPERVISOR	TECHNICIAN	REGISTRAR	DEVOPS	SYS ADMIN
Domain Analysis & Requirements	C	A	C	A	R	I
Ticket Lodging & Photo Upload	R	I	I	I	I	I
Triage, Priority & Tech Dispatch	I	R	I	A	I	I
Physical Maintenance & Job Execution	I	A	R	I	I	I
OTP Sign-off & Verification	R	A	R	I	I	I
SLA Breach Escalation Management	I	R	I	A	I	A
Monthly Maintenance Audit Export	I	C	I	R	I	A
CI/CD Pipeline & Infrastructure Ops	I	I	I	I	R	A

Legend: **R** = Responsible (executes the task); **A** = Accountable (approves outcome); **C** = Consulted (provides input); **I** = Informed (receives notification).

12.2 Communication Governance & Escalation Protocol

- **Level 1 (Supervisor Alert):** Ticket unassigned for > 2 hours triggers automated SMS alert to the Estate Supervisor.
- **Level 2 (Registrar Escalation):** High-severity ticket unresolved after 12 hours triggers automated high-priority email alert to the Campus Registrar.

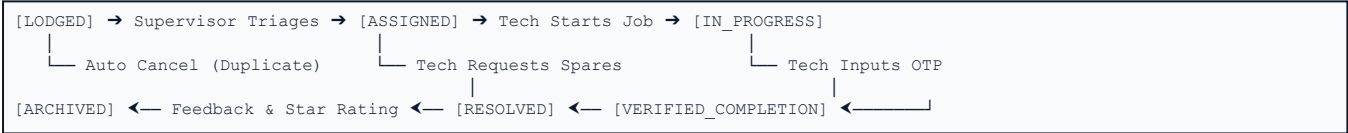
13. FUNCTIONAL REQUIREMENTS SPECIFICATION (FRS)

Functional requirements establish precise software behaviors, transaction boundaries, and automated test pass criteria across the complaint lifecycle.

13.1 Functional Requirements Matrix

ID	REQUIREMENT TITLE	FUNCTIONAL SPECIFICATION & ACCEPTANCE TEST CONDITION
FR-01	Role-Based Authentication	System authenticates users via Bcrypt salted hashes and issues signed JWT tokens. Access is segregated into Student, Supervisor, Technician, and Executive roles.
FR-02	Smart Ticket Lodging	Complainants submit complaints by selecting zone, room, trade category, and uploading photo evidence. System auto-assigns unique ticket ID and priority score.
FR-03	Supervisor Dispatch Kanban	Supervisors view unassigned tickets, filter by trade specialization, and assign jobs to on-duty technicians with an atomic database state update.
FR-04	Mobile Job Card Execution	Technicians view assigned tasks, log consumed spare materials, attach resolution photos, and request the complainant's OTP to close the work order.
FR-05	OTP-Verified Closure	System rejects ticket resolution until the technician provides the 4-digit code generated exclusively on the complainant's active dashboard.
FR-06	Automated SLA Escalation	Background daemon monitors ticket age against SLA thresholds. Breached tickets transition to Escalated and trigger alerts to administrative leadership.
FR-07	Spare Inventory Tracking	System logs spare parts consumed during repair, updating real-time inventory balances and generating purchase requisition alerts when stock drops below threshold.
FR-08	Compliance & Audit Export	Administrators generate aggregated maintenance metrics (MTTR, resolution rates, cost per block) formatted as downloadable CSV and PDF reports.

13.2 Ticket Lifecycle Workflow State Machine



14. ROLE-BASED ACCESS CONTROL (RBAC) & SECURITY

To maintain data privacy, facility accountability, and prevent unauthorized ticket manipulation, the platform enforces strict Role-Based Access Control (RBAC) at the API gateway layer.

14.1 Fine-Grained Security Permission Matrix

PLATFORM OPERATION	STUDENT / RESIDENT	FACILITY SUPERVISOR	FIELD TECHNICIAN	REGISTRAR / EXEC	SYS ADMIN
Lodge Maintenance Ticket	✓ Full	✓ Full	✗ Denied	✗ Denied	✓ Full
View Ticket Status & OTP	✓ Own Only	✓ All Campus	✓ Assigned Only	✓ All Campus	✓ Full
Triage & Assign Technician	✗ Denied	✓ Full	✗ Denied	✗ Denied	✓ Full
Update Job Card & Add Spares	✗ Denied	✓ Override	✓ Assigned Only	✗ Denied	✓ Full
Verify Resolution with OTP	✓ Holds OTP	✗ Denied	✓ Submits OTP	✗ Denied	✓ Override
Re-open Unresolved Ticket	✓ Within 24h	✓ Full	✗ Denied	✓ Full	✓ Full
Export SLA Compliance Reports	✗ Denied	✓ Trade Only	✗ Denied	✓ Full Campus	✓ Full Campus
Manage Users, Roles & Spares	✗ Denied	✗ Denied	✗ Denied	✗ Denied	✓ Full

14.2 Authorization Architecture & Middleware Enforcers

- **Stateless JWT Claims:** Bearer tokens encode `user_id`, `role`, and `zone_id`, verified cryptographically on every API invocation.
- **Horizontal Tenant Scoping:** Students can only retrieve tickets linked to their unique `user_id`, preventing cross-room privacy breaches.
- **Field-Level Immutability:** Once a ticket is marked `Resolved`, the record locks permanently to prevent retrospective tampering.

15. NON-FUNCTIONAL REQUIREMENTS & RULES

15.1 System Quality Attributes

ATTRIBUTE	DESIGN REQUIREMENT TARGET	VERIFICATION & MEASUREMENT METHOD
Security	Bcrypt password hashing (cost factor ≥ 12), TLS 1.3 encryption, and binary magic-byte inspection on image uploads.	Automated security penetration scanning (OWASP ZAP, <code>npm audit</code>) and unauthorized API probing.
Performance	API response times ≤ 250 ms for ticket list queries under a concurrent load of 100 simultaneous requests.	Load testing via k6 / Apache JMeter; recording 95th percentile response times.
Availability	Target 99.9% uptime with zero single points of failure across academic semesters.	Prometheus health checks and automated Kubernetes pod self-healing.
Data Integrity	Strict foreign key constraints; zero orphaned work orders; atomic inventory decrementing transactions.	PostgreSQL transactional rollback unit tests and automated integrity assertions.

15.2 Institutional SLA Rules & Escalation Thresholds

Service Level Agreement (SLA) Resolution Thresholds:

$$\text{SLA Compliance} = \begin{cases} \text{Met}, & \& \text{if } (t_{\text{resolved}} - t_{\text{lodged}}) \leq \text{SLA Limit} \\ \text{Breached}, & \& \text{otherwise} \end{cases}$$

- Emergency Severity (Gas leak, power failure, fire hazard):** SLA limit is **2 Hours**. Auto-escalates to Estate Head if unassigned in 30 minutes.
- High Severity (Room MCB tripping, overflowing drainage):** SLA limit is **6 Hours**. Auto-escalates to Supervisor if unassigned in 2 hours.
- Medium Severity (Broken fan, leaking tap, study lamp failure):** SLA limit is **24 Hours**.
- Low Severity (Furniture repair, wall painting, squeaky hinges):** SLA limit is **72 Hours**.

16. PRODUCT VISION, MVP SCOPE & RISK MATRIX

16.1 Product Vision Statement

"To provide higher-education institutions with an automated, cloud-native Campus Complaint and Maintenance Management System that eradicates manual paper registers, guarantees rapid technician dispatch, enforces mutual OTP-verified work completion, and provides university leadership with transparent, audit-ready infrastructure telemetry."

16.2 Minimum Viable Product (MVP) Scope vs. Future Releases

CORE SYSTEM CAPABILITY	MVP RELEASE SCOPE (SPRINT 1-4)	FUTURE EXPANSION RELEASES
Authentication	Role-based JWT auth (Resident, Supervisor, Tech, Admin)	Single Sign-On (SSO) with Google Workspace / LDAP
Ticket Management	Photo-evidence ticket lodging with room categorization	AI auto-classification of photos using computer vision
Work Dispatch	Supervisor triage kanban with trade filtering	Automated algorithmic dispatch based on GPS technician proximity
Verification	Complainant OTP verification on mobile job cards	NFC tap tags installed inside hostel rooms for physical presence
SLA & Reporting	SLA breach tracking and flat CSV/PDF audit export	Real-time Grafana telemetry dashboards and heatmaps

16.3 Project Risk Management and Mitigation Matrix

IDENTIFIED PROJECT RISK	SEVERITY	PROACTIVE ENGINEERING MITIGATION
Spam or Fake Complaint Flooding	High	Rate-limit lodging API to 3 tickets per resident per hour; enforce mandatory photo upload.
Low Technician App Adoption	High	Build UI with large touch targets, high contrast, minimal typing, and local language support.
Scope Creep Delaying Delivery	Medium	Enforce strict MVP boundaries; defer IoT sensors and smart meters to post-MVP sprints.
Database Lockups on High Concurrency	Medium	Implement composite B-Tree indexes on (status, trade_category, zone_id) tuples.

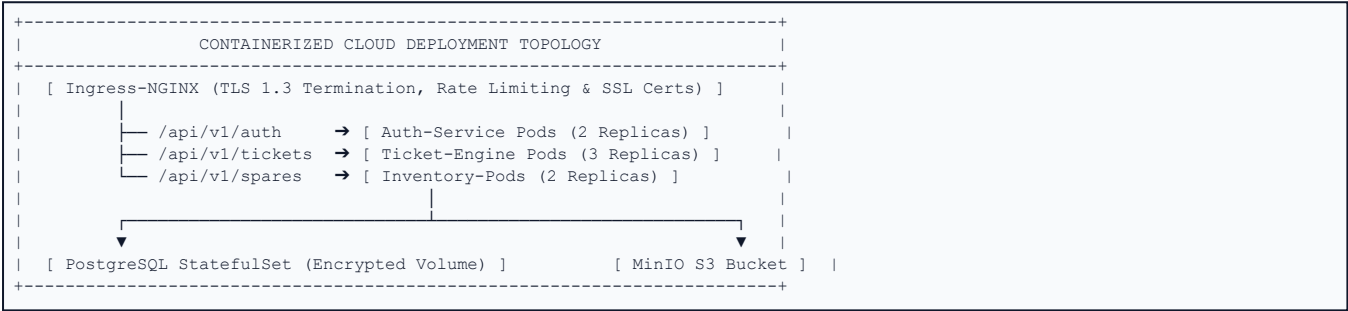
17. SYSTEM ARCHITECTURE & INFRASTRUCTURE

17.1 Logical Multi-Tier System Architecture



17.2 Containerized Cloud DevOps Topology (Future Progression)

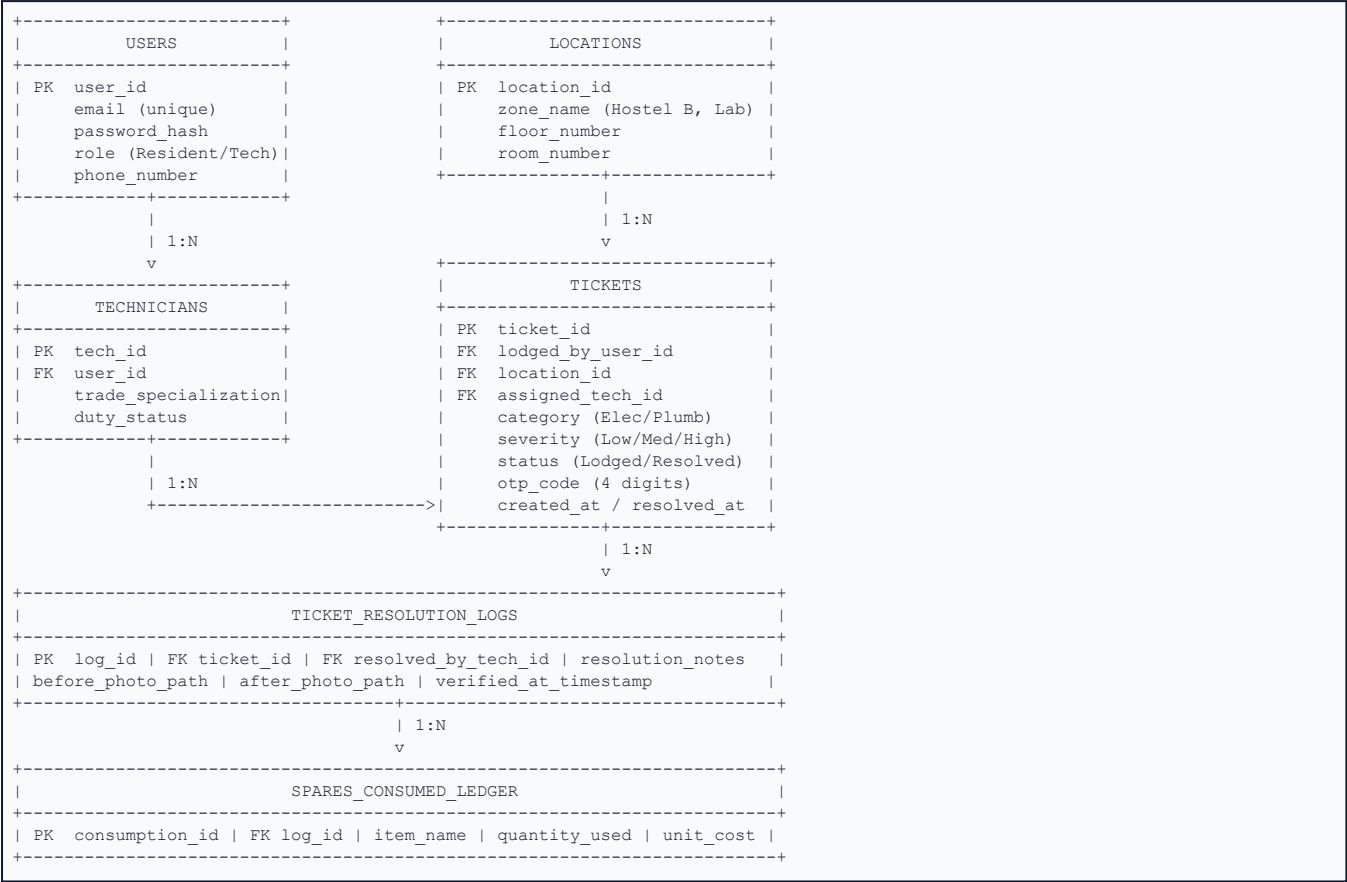
While Unit 1 establishes the design foundation, the system is architected for containerized deployment:



Decoupled Media Storage: Photographic evidence is never stored as binary data in the relational database. Images are compressed client-side, uploaded to private S3 buckets, and referenced by cryptographic hash keys.

18. RELATIONAL SCHEMA & DATA INTEGRITY

18.1 Relational Entity Relationship (ER) Schema Architecture



18.2 Data Dictionary & Integrity Constraints

- USERS.email: Normalized lowercase, RFC 5322 compliant, unique constraint.
- TICKETS.otp_code: Encrypted 4-digit token; cleared from memory immediately upon verified closure.
- TICKETS: Foreign keys enforce ON DELETE RESTRICT to prevent accidental audit record deletion.

19. AGILE ENGINEERING PLAN & ROADMAP

The system engineering lifecycle is organized across four two-week agile sprints, establishing an iterative delivery rhythm from requirements to deployment readiness.

19.1 Four-Sprint Agile Execution Plan

AGILE SPRINT	CORE ENGINEERING DELIVERABLES	REVIEW MILESTONE TARGET
Sprint 1	PostgreSQL schema initialization, migration scripts, JWT role authentication, and student/technician profile models.	Secured authentication system with role separation.
Sprint 2	Complaint lodging module, photo upload pipeline to MinIO object storage, priority scoring algorithm, and ticket state models.	Working student lodging portal with media persistence.
Sprint 3	Supervisor dispatch Kanban, mobile Job Card interface for technicians, OTP validation engine, and closure workflows.	Complete triage, repair, and verification lifecycle.
Sprint 4	SLA breach escalation worker, Registrar audit export engine (CSV/PDF), comprehensive integration test suite, and MVP review.	Production-ready MVP verified against test plan.

19.2 Multi-Phase Course Progression Roadmap



Unit 1 Scope Boundary: Unit 1 establishes the requirements specification, wireframes, architecture, and Git repository structure. Container packaging (Docker), Infrastructure-as-Code (Terraform/Ansible), Kubernetes manifests, and Prometheus telemetry will be implemented during subsequent course units.

20. GIT REPOSITORY ARCHITECTURE & SETUP

20.1 Standardized Directory Hierarchy

```
campus-complaint-management-system/
├── .github/workflows/      # CI/CD pipelines (GitHub Actions)
├── app/
│   ├── modules/           # auth/, tickets/, dispatch/, inventory/, reports/
│   ├── database/          # migrations/ & models/
│   └── tests/             # unit/ & integration/
├── docs/                  # Architecture diagrams, wireframes & BRD
├── docker/               # Dockerfiles & docker-compose configurations
├── kubernetes/           # Deployment manifests, services, and ingress
├── terraform/            # Cloud infrastructure automation scripts
├── ansible/              # Server configuration management playbooks
├── monitoring/           # Prometheus alerting & Grafana dashboards
└── README.md             # System onboarding and setup instructions
```

20.2 PowerShell / Bash Initialization Script

Execute the following commands to initialize the standardized project layout and Git tracking:

```
# Initialize local Git repository
mkdir campus-complaint-management-system
cd campus-complaint-management-system
git init

# Construct enterprise project directory structure
mkdir -p .github/workflows
mkdir -p app/modules/auth app/modules/tickets app/modules/dispatch app/modules/inventory app/modules/reports
mkdir -p app/database/migrations app/database/models
mkdir -p app/tests/unit app/tests/integration
mkdir -p docs docker kubernetes terraform ansible monitoring

# Initialize placeholder documentation
touch app/modules/README.md app/database/README.md app/tests/README.md
touch docs/README.md docker/README.md kubernetes/README.md
echo "# Campus Complaint & Maintenance Management System" > README.md

# Stage and commit baseline structure
git add .
git commit -m "feat: initialize Unit 1 project directory structure and repo layout"
git status
```

21. VERIFICATION STRATEGY & TEST PLAN

The verification strategy defines structured functional scenarios designed to validate application logic, role boundaries, and data integrity during subsequent implementation sprints.

21.1 Acceptance Test Scenarios Matrix

ID	VERIFICATION SCENARIO	EXPECTED OPERATIONAL BEHAVIOR & ACCEPTANCE CRITERIA	STATUS
TC-01	Valid and invalid user credentials	Correct credentials return a signed JWT token; invalid passwords return HTTP 401 with descriptive error.	Planned
TC-02	Cross-role privilege isolation	A resident attempting to access supervisor dispatch endpoints receives an HTTP 403 Forbidden.	Planned
TC-03	Photo upload MIME validation	System inspects binary magic bytes; rejects .exe or script files disguised as images.	Planned
TC-04	Invalid OTP ticket closure	Technician entering an incorrect OTP is blocked from closing the ticket; logs failed attempt.	Planned
TC-05	Atomic technician dispatch	Concurrent assignment requests to the same technician execute atomically without race conditions.	Planned
TC-06	SLA timer background escalation	Tickets unassigned past SLA limit transition to <code>Escalated</code> state and send alert emails.	Planned
TC-07	Maintenance ledger CSV export	Generated CSV contains all mandatory institutional audit fields (Ticket ID, Zone, Tech, Spares, MTTR).	Planned
TC-08	Database disaster recovery	PostgreSQL backup restores with zero orphaned logs and complete transactional integrity.	Planned

21.2 Suggested Demonstration Walkthrough Sequence

- Student Lodging:** Log in as Shivraj Jadhav (2126UDSM2588), lodge a sparking geyser complaint with photo, verify unique ID, and observe OTP generated.
- Supervisor Dispatch:** Log in as Ramesh Pawar, inspect the triage kanban, assess high severity, and dispatch electrician Mahesh Shinde.
- Technician Closure:** Log in as Mahesh Shinde on mobile, enter consumed spare parts, upload resolution photo, input student OTP 7491, and verify ticket marked resolved.
- Executive Telemetry:** Log in as Registrar Dr. Joshi, review updated campus MTTR telemetry, and export the monthly compliance audit.

22. DELIVERABLES CHECKLIST & DEFENSE GUIDE

22.1 Deliverables Traceability Matrix

COURSE SYLLABUS DELIVERABLE	REPORT COVERAGE	ACADEMIC STATUS
Business Problem Statement & Justification Report	Section 1 & Section 2	✓ Complete
Product Vision Statement & MVP Feature Matrix	Section 16	✓ Complete
Stakeholder Responsibility Matrix & Personas	Section 11 & Section 12	✓ Complete
Functional (FRS) and Non-Functional (NFRS) Specs	Section 13, 14 & 15	✓ Complete
Design Thinking Report & Low-Fidelity Prototypes	Sections 3, 4, 5, 6, 7, 8 & 9	✓ Complete
IBM Design Thinking & IBM Loop Execution Specs	Section 10	✓ Complete
Multi-Phase Product Engineering Roadmap	Section 19	✓ Complete
System Architecture & Relational Database Schema	Section 17 & Section 18	✓ Complete
Initialized Git Repository & Standardized Folder Layout	Section 20	✓ Complete (Scripts Documented)
Verification Strategy & Acceptance Test Plan	Section 21	✓ Complete

22.2 Review Presentation Sequence for Evaluators

When presenting this Unit 1 milestone before institutional examiners, follow this 5-step narrative structure:

- The Campus Operational Bottleneck:** Articulate the chaos universities face with physical complaint registers, untracked repairs, absent SLAs, and unverified verbal technician updates.
- Design Thinking Evolution:** Walk through *Empathize* → *Define* → *Ideate* → *Prototype* → *Test*, demonstrating how student and technician friction shaped the UI wireframes.
- IBM Loop Application:** Explain how *Observe* → *Reflect* → *Make* resolved fake ticket closures by introducing mutual OTP-verification.
- Architectural Integrity:** Present the multi-tier architecture and relational database schema, highlighting the tamper-evident `TICKET_RESOLUTION_LOGS` table and RBAC rules.
- Agile Roadmap & Git Layout:** Present the 4-sprint development plan and the enterprise repository architecture established for future CI/CD, Docker, and Kubernetes units.